

Cintara Senior Engineer Technical Exercise

Purpose

Thank you for taking the time to complete this exercise.

This exercise is designed to help us understand how you think about software design, tradeoffs, and implementation. We are not looking for a perfect solution, nor are we evaluating based on lines of code.

Please optimize for clarity and simplicity.

You may use any programming language, framework, editor, AI coding assistant, or development workflow that you normally use.

Time Expectation

Please spend no more than **2 hours** on this exercise.

We understand actual time spent may vary, but we encourage you to treat this as a time-boxed exercise and make pragmatic tradeoffs.

Problem Statement

Build a minimal workflow execution service.

A workflow consists of a sequence of steps.

Each step may receive input and produce output that can be consumed by subsequent steps.

Example workflow definition:

```
{
  "workflowId": "contract-review",
  "steps": [
    {
      "name": "validate",
      "input": {
        "contractId": "123"
      }
    }
  ]
}
```

```

    },
    {
      "name": "approve"
    },
    {
      "name": "execute"
    }
  ]
}

```

Example execution result:

```

{
  "executionId": "abc123",
  "status": "completed",
  "steps": [
    {
      "name": "validate",
      "status": "completed",
      "output": {
        "valid": true
      }
    },
    {
      "name": "approve",
      "status": "completed",
      "output": {
        "approvedBy": "system"
      }
    },
    {
      "name": "execute",
      "status": "completed",
      "output": {
        "result": "success"
      }
    }
  ]
}

```

Functional Requirements

Your system should support:

Create Workflow

Store a workflow definition.

Execute Workflow

Execute workflow steps in order.

A step may produce output that becomes available to subsequent steps.

You may choose how workflow context and step outputs are represented.

Track Execution State

Provide visibility into execution progress.

Example:

```
{
  "executionId": "123",
  "status": "running",
  "currentStep": 2
}
```

Execution History

Return execution history for a workflow execution.

Include step-level execution information.

Requirements

- Expose APIs for interacting with the system.
 - Persistence may be in-memory or durable.
 - Make reasonable assumptions where necessary.
 - Document assumptions in the README.
 - Optimize for simplicity and clarity.
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Deliverables

Please submit:

1. Source code
2. README
3. Instructions for running locally

The README should include:

- Architecture overview
 - Key assumptions
 - Tradeoffs made
 - What you would improve with an additional week
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Guidance

We are intentionally leaving several aspects of the system open-ended.

There is no single correct solution.

We are interested in understanding:

- Design choices
- Tradeoffs
- Assumptions
- Evolution of the system

During the interview we will discuss how your design would evolve under changing requirements.

Submission

Please provide a link to a Git repository containing your solution and supporting documentation.

Thank you and we look forward to the discussion.